Phase 1: Understanding the Existing 8-bit Discrete SAR ADC

Complete explanations and questions discussed

Project: ELT10250 ADC Board for Educational Use
Reference: K24007 Discrete ADC Board documentation (2026)
Scope: Original circuit and preparation for the redesign

1 Purpose of Phase 1

The university assignment asks us to reengineer an existing **8-bit successive-approximation-register (SAR) ADC** for educational use. The original board uses individual logic integrated circuits (ICs), rather than a dedicated ADC IC. The intended redesign adds an external clock input (`CLK_IN`), an external analog input from a waveform generator, and measurement points for relevant signals.

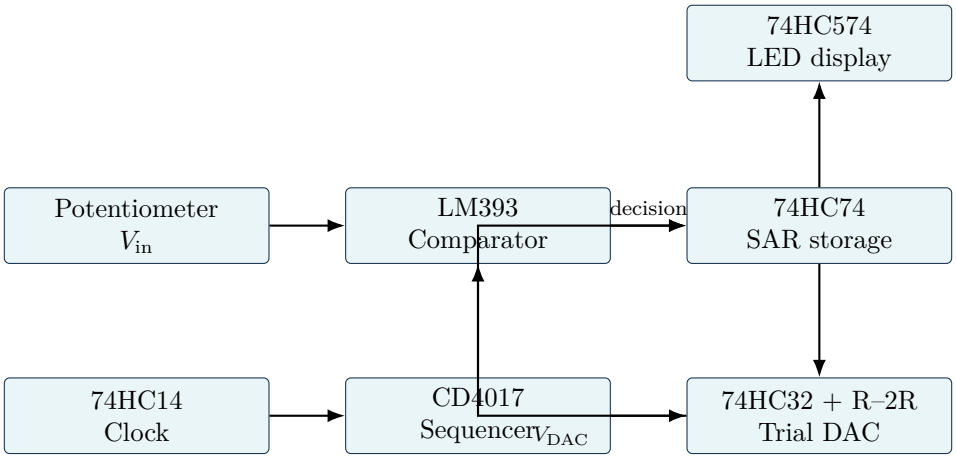
The goal of Phase 1 is to explain *what each electronic component does*, *which signals it receives*, *what it produces*, and *how the next block uses its output*. This document also records the questions that arose during our discussion and their answers.

2 Overview: what is an 8-bit SAR ADC?

An analog-to-digital converter converts a continuous voltage into a digital number. For a nominal 0–5 V input range, an 8-bit result has $2^8 = 256$ possible codes, numbered 0 through 255. The ideal size of one code step is

$$\Delta V = \frac{5\,\mathrm{V}}{256} = 0.01953125\,\mathrm{V} \approx 19.53\,\mathrm{mV}. \tag{1}$$

A SAR ADC conducts a binary search. It first tries the most significant bit (MSB, bit 7), then bit 6, and continues to bit 0. In each trial, a DAC produces an analog guess, a comparator decides if the guess is above or below the input, and the SAR register remembers the answer.



Conceptual connections: actual gate wiring and timing details are more complicated. The LM7805 supplies power throughout the digital circuit.

3 Block 1: LM7805 power supply

3.1 Purpose and components

The board receives about 9–12 V DC through a barrel connector (X1). Its 74HC digital logic is powered from a regulated 5 V rail, so the higher adapter voltage must not be connected directly to those ICs.

$$\boxed{\text{9–12 V DC}} \longrightarrow \boxed{\text{D1: 1N4001}} \longrightarrow \boxed{\text{IC1: LM7805}} \longrightarrow \boxed{\text{5 V rail}}$$

- **X1 barrel connector:** brings the external supply and ground onto the PCB.
- **D1, 1N4001 diode:** series reverse-polarity protection; with correct polarity, it has a forward voltage drop (often roughly 0.6–0.9 V).

- **LM7805:** a linear regulator maintaining approximately 5 V at its output.
- **Input/output capacitors:** reduce disturbances and support stable regulation. Bulk electrolytic capacitors help with slower changes; small ceramic capacitors bypass faster noise.
- **GND:** common zero-volt reference and current return path.

3.2 Important regulator pins and operation

For the standard front-facing TO-220 orientation (printed face toward the viewer, leads pointing down):

Pin	Name	Role
1	IN	Voltage after the protection diode
2	GND	Common ground
3	OUT	Regulated 5 V

The regulator uses internal feedback to correct changes in its output voltage. Being a *linear* regulator, it mainly dissipates the excess voltage as heat:

$$P_{\text{reg}} \approx (V_{\text{reg,in}} - 5\,\text{V})I_{\text{load}}. \tag{2}$$

For example, if the regulator receives 11.3 V after D1 and supplies 0.10 A, it dissipates approximately $(11.3 - 5) \times 0.10 = 0.63\,\text{W}$.

Key idea: The LM7805 provides power; it does not generate clock pulses or perform ADC comparisons. The reference documentation states that the LM393 comparator instead uses the higher, unregulated supply for input-voltage headroom, while its digital output is pulled up to 5 V.

4 Block 2: 74HC14 clock generator

4.1 What is a clock?

A clock is a repeating digital signal, ideally alternating between LOW (near 0 V) and HIGH (near 5 V). The CD4017 advances to its next state on a valid rising clock edge. The clock therefore controls *when* successive approximation operations occur.

4.2 What is a Schmitt-trigger inverter?

The 74HC14 contains six inverting Schmitt-trigger gates. Inversion means a low input produces a high output, and a high input produces a low output. Unlike a simple inverter, a Schmitt-trigger inverter has **two switching thresholds**:

- **Upper threshold V_{T+}:** if the input voltage is *rising*, the output switches HIGH → LOW when the upper threshold is crossed.
- **Lower threshold V_{T-}:** if the input voltage is *falling*, the output switches LOW → HIGH when the lower threshold is crossed.

The reference PDF reports approximately $V_{T+} = 2.56\,\text{V}$ and $V_{T-} = 1.80\,\text{V}$ for its illustrated conditions. Their difference is the hysteresis width:

$$V_H = V_{T+} - V_{T-} = 2.56 - 1.80 = 0.76\,\text{V}. \tag{3}$$

The exact thresholds are IC- and condition-dependent, *not universal fixed values* for all 74HC14 chips.

4.3 Our doubt: why are upper and lower thresholds needed?

With only one switching threshold, slow or noisy input variations near that threshold can create multiple unwanted output transitions. Two thresholds make the output *remember its previous state* while the input is between 1.80 and 2.56 V. For example, at 2.20 V the output may be HIGH if the input was rising from below, or LOW if it was falling from above.

Our question: Why exactly 1.80 V and 2.56 V? They result from the 74HC14's internal transistor design and operating conditions. They are not set by our external RC values, and a different Schmitt-trigger design could have different thresholds.

4.4 How the RC oscillator works

The first Schmitt-trigger inverter is connected in a feedback arrangement with a resistor R1 and timing capacitor C:

1. On startup the capacitor is initially low; the inverter output goes HIGH.
2. The output charges the capacitor through R1.
3. When capacitor voltage reaches V_{T+}, the inverter output switches LOW.
4. The capacitor discharges through R1 toward the low output.
5. At V_{T-}, the output switches HIGH again.
6. The charge/discharge cycle repeats and produces a square-wave clock.

The next 74HC14 inverter in the Figure 3.3 schematic buffers and inverts the oscillator signal to provide a clock output.

Our doubt: Does the Schmitt trigger alone determine frequency? No. The resistor and capacitor mainly set the time scale, and the thresholds also influence the period:

$$\tau = RC, \qquad f \propto \frac{1}{RC} \quad \text{(for fixed thresholds and supply).} \tag{4}$$

Increasing R or C generally decreases the oscillator frequency. The Schmitt trigger provides the repeatable switching action, but the *RC feedback* is necessary for this oscillator.

4.5 Our doubt: is the existing clock fixed, and do we need another?

The original clock is an internal RC oscillator with a fixed *nominal* frequency, although its actual frequency varies with tolerances and conditions. The university assignment explicitly calls for an external adjustable `CLK_IN` driven by a waveform generator. We can either replace the internal clock path or retain both using a carefully designed selection and input-conditioning circuit. Keeping both is a proposed enhancement, not an explicit requirement of the assignment.

4.6 Important discrepancy in the reference PDF

The source is not internally consistent on the clock values:

- Figure 3.3 on PDF page 12 labels an oscillator capacitor **C10 = 100 nF** and R1 = 1 kΩ, and annotates the clock at roughly **16 kHz**.
- The accompanying prose refers to **C19** as the timing capacitor.
- The bill of materials lists **C19 = 100 pF**.
- The testing chapter expects **70–130 Hz**, with roughly ten conversions per second.

These are conflicting statements, not verified component values for one functioning design. **Do not copy the oscillator values without checking the actual intended schematic and calculating or measuring the clock frequency.**

5 Block 3: CD4017 SAR sequencer

5.1 What is a decade counter?

The CD4017 is a decade counter with ten *decoded* outputs, Q0 through Q9. It internally stores its current counting state and makes exactly one of these outputs HIGH during normal stable operation. At each permitted rising clock edge it advances to the next state.

Our doubt: Does it store values? Yes, it stores its *current position* (for example, "currently Q3"), but **not** the final eight-bit ADC conversion result. That result is stored by the 74HC74 flip-flops and then captured by the 74HC574.

Our understanding: "It tells which bit is tested." Correct. The CD4017 is the sequence controller: it selects the current approximation stage, not the 0/1 comparison result.

5.2 Important CD4017 pins

Pin	Signal	Function
14	CLK	Clock input
13	Clock inhibit	HIGH inhibits counting
15	RESET	HIGH forces Q0 active
16	VDD	+5 V supply on this board
8	GND	Ground
3,2,4,7,10,1,5,6,9,11	Q0–Q9	Ten sequential control outputs

5.3 Sequence described by the reference PDF

Active output	Intended operation
Q0	Begin trial of bit 7 (MSB).
Q1	Capture decision for bit 7; trial bit 6.
Q2	Capture bit 6; trial bit 5.
Q3	Capture bit 5; trial bit 4.
Q4	Capture bit 4; trial bit 3.
Q5	Capture bit 3; trial bit 2.
Q6	Capture bit 2; trial bit 1.
Q7	Capture bit 1; trial bit 0 (LSB).
Q8	Reference describes capturing final bit 0 and output result.
Q9	Clear SAR flip-flops and reset counter to Q0.

The text states that Q9 participates in resetting the CD4017. A reset caused by Q9 can shorten the Q9 pulse, because RESET immediately forces the count back to Q0.

6 Block 4: 74HC74 SAR storage

6.1 Why four ICs instead of eight?

The 74HC74 contains **two** D-type flip-flops per IC. A single D flip-flop stores one bit. Therefore the reference board needs

$$4 \text{ ICs} \times 2 \text{ flip-flops/IC} = 8 \text{ bit storage elements.} \tag{5}$$

The four devices are IC5–IC8. **Our doubt: "Are there eight 74HC74 chips?"** No: there are four physical 74HC74 chips containing eight flip-flops in total.

6.2 How a D flip-flop works

D is the data input and **Q** is the stored output. On a valid rising clock edge, Q captures the value of D. Between edges, Q remains stored (unless asynchronous preset or clear changes it).

D before rising edge	Rising clock edge?	Q afterward
1	Yes	1
0	Yes	0
D changes	No	Previous Q remains

Each flip-flop also has active-LOW **CLR** (force Q = 0) and **PRE** (force Q = 1). These asynchronous controls do not require a clock edge. Do not assert PRE and CLR LOW simultaneously as a normal operating condition.

6.3 Important pins, first half of one 74HC74

Pin	Signal	Role
2	D	Data input
3	CLK	Rising-edge capture
5	Q	Stored bit
6	$\overline{Q}$	Complementary output
1	$\overline{\text{CLR}}$	Active-LOW clear
4	$\overline{\text{PRE}}$	Active-LOW preset
14 / 7	VCC / GND	Power

The second flip-flop is on pins 12 (D), 11 (CLK), 9 (Q), 8 ($\overline{Q}$), 13 ($\overline{\text{CLR}}$), and 10 ($\overline{\text{PRE}}$).

6.4 Role during conversion

The comparator produces a binary decision about the current DAC trial. At the appropriate capture event, a 74HC74 flip-flop stores that decision for one bit. The stored bits remain available while later, less significant bits are tested. Q outputs feed the trial DAC logic and the output latch.

7 Block 5: 74HC32 OR gates and R–2R DAC

7.1 Why is there a DAC in an ADC?

The LM393 comparator needs two *analog* voltages. The input potentiometer provides V_{in}, while the DAC turns the temporary binary trial code into an analog guess V_{DAC}.

7.2 What does OR mean?

A 74HC32 contains four two-input OR gates. Two ICs provide eight gates. An OR gate produces a 1 if *either* input (or both) is 1:

Stored bit A	Trial bit B	DAC bit $A \vee B$
0	0	0
0	1	1
1	0	1
1	1	1

7.3 Our doubt: why do we need OR gates?

The DAC must temporarily assume the bit under test is 1, *before* knowing whether the comparator will accept it. At the same time, all previously accepted bits must remain 1. The OR gate combines two sources of 1:

$$\text{DAC bit} = \text{previously stored bit} \ \vee \ \text{current trial signal}. \tag{6}$$

Example: stored bits `10000000` (bit 7 accepted), while the CD4017 selects trial bit 6, `01000000`. Bitwise OR yields `11000000`, and the DAC tests 3.75 V in the ideal 5 V model.

If bit 6 is **accepted**, its flip-flop stores 1. After the trial signal disappears, $1 \vee 0 = 1$: the bit stays high. If bit 6 is **rejected**, its flip-flop stores 0. After the trial signal disappears, $0 \vee 0 = 0$: the bit returns low.

Key idea: The OR gate does not make the comparison decision. It permits a temporary test bit to be asserted while preserving bits already accepted by the SAR register.

7.4 Why an R–2R ladder?

The ladder uses resistance values in the ratio $R : 2R$, which are approximately 10 kΩ and 20 kΩ in the reference BOM. The circuit gives each bit half the weight of its more significant neighbor. In an *ideal, unloaded*, eight-bit, 5 V DAC,

$$V_{\text{DAC}} = V_{\text{ref}} \frac{D}{2^8} = 5\,\text{V} \frac{D}{256}, \quad 0 \le D \le 255. \tag{7}$$

Bit	Weight	Ideal voltage if set
7	128	2.5000 V
6	64	1.2500 V
5	32	0.6250 V
4	16	0.3125 V
3	8	0.15625 V
2	4	0.078125 V
1	2	0.0390625 V
0	1	0.01953125 V

The maximum ideal code 255 corresponds to $255/256 \times 5\,\text{V} \approx 4.9805\,\text{V}$, not exactly 5 V. Real output voltages depend on resistor tolerance, 74HC32 output levels, and output loading. Figure 3.6 should be checked when finalizing the exact ladder connections.

8 Block 6: LM393 comparator

8.1 Purpose, inputs, and output

The LM393 compares the measured analog voltage to the DAC trial voltage. The reference uses one of its two comparator channels:

Pin	Function	Board signal
3	Non-inverting (+) input	V_{in} from potentiometer
2	Inverting (−) input	V_{DAC} from ladder
1	Comparator output	Decision to SAR logic
8	Positive supply	Higher unregulated input rail
4	GND	Ground

The intended logic is

$$\begin{cases} V_{\text{in}} > V_{\text{DAC}} & \Rightarrow \text{HIGH decision; keep the trial bit,} \\ V_{\text{in}} < V_{\text{DAC}} & \Rightarrow \text{LOW decision; reject the trial bit.} \end{cases} \tag{8}$$

Near equality, comparator offset and noise can affect the outcome; the PDF's basic higher/lower description does not specify an exact equality rule.

8.2 Our doubt: why is a pull-up resistor necessary?

The LM393 has an **open-collector output**: it can actively pull its output LOW, but it cannot actively drive it HIGH. R4 (10 kΩ) connects the output to the regulated +5 V rail. When the internal output transistor is OFF, R4 pulls the signal HIGH; when the transistor is ON, the signal is pulled LOW. The resistor also limits current (approximately $5\,\text{V}/10\,\text{k}\Omega = 0.5\,\text{mA}$ in the simplified LOW-state model).

8.3 Why is the LM393 not powered from the 5 V rail?

The reference PDF explains that the LM393 input common-mode voltage range cannot normally reach its positive supply rail. Supplying it from the higher raw input rail allows its inputs to accept the intended near-0–5 V analog signals. The output pull-up is still connected to 5 V to maintain safe logic levels for the 74HC ICs.

9 Block 7: 74HC574 output register and LEDs

The 74HC574 is one chip containing **eight D flip-flops**. It captures all eight data inputs together at its active clock edge, then holds that completed result for the LEDs. Its output enable (OE) must be active (LOW) for the output pins to drive the display. Each LED has a current-limiting resistor (R21–R28, 470 Ω in the reference BOM).

Our doubt: "Is one 74HC574 enough for full storage?" Yes. There are **four 74HC74 chips** containing eight working flip-flops and **one 74HC574 chip** containing eight output flip-flops.

$$\boxed{4 \times \text{74HC74} = \text{working 8-bit SAR register}} \longrightarrow$$

$$\boxed{1 \times \text{74HC574} = \text{stable 8-bit output latch}} \longrightarrow \boxed{8\ \text{LEDs}}$$

This division matters because the SAR bits change during each approximation. The output latch keeps the previous completed result visible while the next conversion is underway.

10 Worked example: converting a 3.0 V input

Assume an *ideal* $V_{\text{ref}} = 5\,\text{V}$ R–2R DAC and a stable $V_{\text{in}} = 3.0\,\text{V}$. The following table illustrates the binary-search decisions and does not attempt to represent every gate propagation delay:

Tested bit	DAC trial bits	DAC voltage	Comparator decision
7	`10000000`	2.5000 V	Keep (1)
6	`11000000`	3.7500 V	Reject (0)
5	`10100000`	3.1250 V	Reject (0)
4	`10010000`	2.8125 V	Keep (1)
3	`10011000`	2.9688 V	Keep (1)
2	`10011100`	3.0469 V	Reject (0)
1	`10011010`	3.0078 V	Reject (0)
0	`10011001`	2.9883 V	Keep (1)

The final code is

$$\texttt{10011001}_2 = 128 + 16 + 8 + 1 = 153_{10}, \quad V_{\mathrm{DAC}} = \frac{153}{256} \times 5\,\mathrm{V} \approx 2.9883\,\mathrm{V}. \tag{9}$$

The 74HC574 then captures the eight-bit result and updates the LEDs.

11 Critical design checks before reproducing the PCB

1. **Clock component and frequency discrepancy:** as noted above, Figure 3.3, prose, BOM, and test instructions disagree. The expected 70–130 Hz test frequency must not be assumed to follow from all stated R/C values.
2. **Output-latch timing:** Section 3.4.1 describes Q8 as both capturing the final SAR bit and capturing the output in the 74HC574. A flip-flop's Q output changes only after a propagation delay. If both devices sample on the same edge, the 74HC574 may capture the prior value of the least significant bit. The actual schematic's clock polarity, wiring and timing need independent verification before copying it.
3. **Input source:** the original potentiometer is slow-moving. The PDF notes that the board has no sample-and-hold circuit; a rapidly changing external waveform could therefore change during a conversion and corrupt the result.
4. **External clock:** the ELT10250 specification requires an external waveform-generator clock input. Design appropriate voltage levels, reference ground, protection/conditioning and source selection. Do not simply short an external generator output to the internal oscillator output.
5. **Test points:** useful candidates include +5 V, GND, `CLK_IN`, CD4017 Q stages, V_{in}, V_{DAC}, LM393 output, SAR bit outputs, output-latch timing, and reset signals. Final test-point selection needs schematic and PCB review.

12 Answers to our key questions (quick revision)

Question we asked	Answer to remember
Why does the 74HC14 have two thresholds?	To provide hysteresis: the output does not chatter with small noisy input changes near one threshold.
Why 1.80 V and 2.56 V?	These are approximate switching thresholds reported for the reference's conditions, arising from the IC's internal circuit.
Do R and C make the Schmitt trigger oscillate?	R and C create the time-varying voltage through feedback; the Schmitt trigger switches at its thresholds. Together they form the oscillator.
Does the Schmitt trigger alone fix the clock frequency?	No. R, C, thresholds and operating conditions affect the frequency.
Do we need an adjustable clock for the new board?	Yes, the assignment asks for an external `CLK_IN`; keeping an internal clock is optional.
Does CD4017 store values?	It stores the current sequence state, not the eight-bit ADC answer.
Does CD4017 tell which bit to test?	Yes. It controls the bit-test sequence.
Are there eight 74HC74 chips?	No: four chips, each with two flip-flops = eight stored SAR bits.
Is one 74HC574 enough for final storage?	Yes: one 74HC574 contains eight flip-flops and holds the display result.

Question we asked	Answer to remember
What is the OR gate doing?	For each bit it combines the already stored bit with the temporary trial signal; either 1 makes the DAC bit 1.
Who decides whether a trial bit is 0 or 1?	The LM393 comparator, based on V_{in} versus V_{DAC}.
Who actually remembers that comparator decision?	A 74HC74 D flip-flop at its appropriate capture event.
Why does LM393 need R4?	Its open-collector output can pull LOW but requires a pull-up to produce HIGH.

13 Phase 1 final explanation for a project meeting

The existing circuit is an eight-bit successive-approximation ADC built from separate logic ICs. An LM7805 supplies regulated 5 V to the digital circuitry. A 74HC14 Schmitt-trigger RC oscillator produces clock pulses. A CD4017 uses the pulses to control which bit is tested, beginning with the most significant bit. Four dual 74HC74 chips store the eight comparator decisions. Eight OR gates from two 74HC32 chips combine stored bits with the active trial bit. A resistor ladder converts this temporary binary code into an analog DAC voltage. The LM393 comparator checks the DAC voltage against the input from the potentiometer and reports whether the current trial is too high or can be retained. After eight trials, a 74HC574 captures the completed binary result and drives eight LEDs. The circuit then resets and repeats.